

Re:Learn

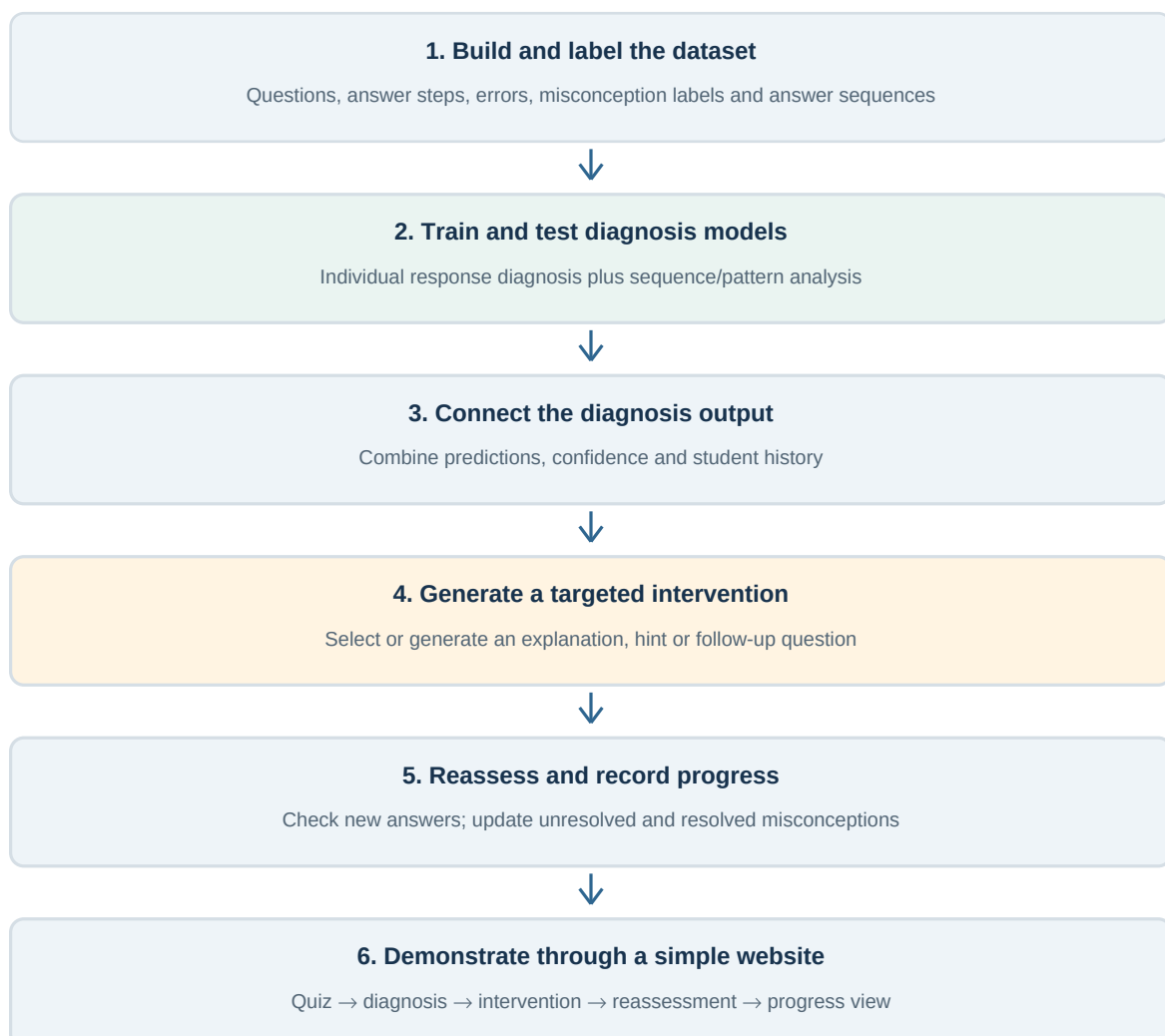
Technical project documentation • Dataset, AI models, website integration and optional teaching features

Project in a few lines

Re:Learn aims to identify **why** a student gives a wrong answer—not only whether the answer is wrong. It analyses individual responses and patterns across related questions, then provides a targeted intervention and checks whether the misunderstanding has improved.

Scope note: This document is based on the problem statement and requirements discussed in this conversation. The central work is misconception diagnosis, personalised intervention, reassessment and evaluation. A full teaching LMS is not the main deliverable; a small website can be used to demonstrate the system.

Recommended end-to-end plan



Part 1 — Dataset: what to build first

Should you use a Class 10 Physics textbook?

Yes. Use a textbook as a **content source** to choose chapters, concepts, definitions, formulas and question ideas. But a textbook alone is not a misconception dataset. You must add realistic wrong answers, the reasoning behind them, and expert-reviewed labels. Check the textbook's usage rights; for a demo, use a permitted source or write original questions aligned to the syllabus.

Recommended dataset structure

Field	What it contains	Example
question_id	Unique question identifier	PHY-MOT-001
topic / concept	Chapter and concept tested	Motion / speed
question + context	Question wording, units and any diagram context	Distance = 100 m; time = 5 s. Find speed.
correct answer + steps	Expected result and valid solution path	$100 \div 5 = 20$ m/s
student response	Wrong answer and, where possible, working or explanation	500 m/s; multiplied distance by time
misconception label	Specific concept error, not just "wrong"	Uses multiplication instead of division
error type / alternative	Calculation slip, unit error, guess, unclear response, etc.	Possible arithmetic slip
confidence / evidence	How certain the label is and what supports it	Medium; working not shown
sequence_id + order	Links responses from one quiz or learning session	Quiz-07, Q1 → Q2 → Q3
sequence-level label	Pattern or connected misconception supported by the full sequence	Repeatedly confuses speed and acceleration

Build two related datasets

- **Individual-response dataset:** one record per response. It teaches the model to classify the likely reason for a specific answer.
- **Sequence dataset:** one record per student attempt or ordered group of related responses. It teaches the system to find patterns across questions and concepts.

A sequence can reveal a pattern that one answer cannot. For example, a student may answer several questions in ways that suggest they treat speed and acceleration as the same thing. Store the order of questions and answers, not just a bag of responses.

How to create reliable labels

- Start with a small concept map for one Physics chapter. List the concepts and common misconceptions for each concept.
- Write original questions that test each concept in more than one way. Include questions where different misconceptions can lead to the same wrong answer.
- Create plausible wrong-answer examples by reasoning through each misconception. Do not invent labels based only on the final number.

- Ask a Physics teacher or knowledgeable reviewer to check the question, answer, misconception label and explanation.
- Include “insufficient evidence”, “careless/calculation error” and “other/unknown” labels. The system must be allowed to say it is unsure.
- Create multi-question quiz examples with known patterns. Keep the full sequence and its reviewed interpretation.
- Split train, validation and test data by question templates or misconception examples—not random near-duplicates—so the test measures generalisation.

Important: The same wrong answer can have multiple causes. If the student's working is unavailable, the model should return likely causes with confidence or ask a short diagnostic follow-up question instead of claiming certainty.

Part 2 — Models and AI components

You do not need a separate large AI model for every feature. Build one integrated system from specialised components. Some components are trained models; others can be ordinary software or an existing language model.

Component / model	Purpose	Train on / build from	Output
A. Individual misconception classifier	Predict the likely misconception from one response and its context.	Individual-response dataset: question, topic, student answer, working, correct answer and reviewed label. Start with a baseline such as TF-IDF + Logistic Regression/SVM or text embeddings + a classifier.	Misconception label(s), confidence, or unsure.
B. Sequence/pattern model	Find recurring or connected misconceptions across ordered answers.	Sequence dataset: ordered questions, responses, per-answer evidence and sequence-level labels. Start with rules/features or a sequence classifier; consider a small Transformer/GRU only if enough data exists.	Pattern-level misconception(s), evidence across questions, confidence.
C. Intervention generator	Explain the diagnosed issue and create a targeted hint, example or follow-up question.	Usually no new model training is required for a prototype. Use an LLM with a controlled prompt, diagnosis result, approved content and output format. Optionally use a curated intervention bank.	Explanation, worked example, hint and next question.
D. Reassessment / mastery logic	Check whether the misconception is still present and update the learner record.	Begin with application logic and reviewed follow-up questions. It need not be a trained AI model. Later, use a knowledge-tracing approach if data supports it.	Resolved / still likely / uncertain; updated progress.
E. Optional speech/visual tools	Speak explanations, animate an avatar or render whiteboard instructions.	Use existing text-to-speech, avatar/animation tools and drawing/rendering code. These are not misconception models.	Audio, avatar motion, diagrams and animation.

Suggested model strategy

- **Build a baseline first.** A simple classifier and transparent rules give you a measurable starting point. Do not begin by training a large model from scratch.
- **Keep individual and sequence tasks distinct.** They use different input shapes and labels. Their predictions can be combined by a diagnosis service.
- **Use an LLM for teaching, not as proof of diagnosis accuracy.** Feed it the diagnosis and vetted learning material. Constrain it to avoid inventing facts or confidently explaining an uncertain cause.
- **Use confidence and abstention.** When evidence is weak or causes are ambiguous, ask a diagnostic question or show more than one possible explanation.

Example: a student gives 500 m/s for 100 m in 5 s. The individual classifier may flag “multiplies instead of divides.” If the student also repeatedly confuses distance, speed and acceleration across the quiz, the sequence component can flag a broader pattern. The intervention engine then selects an explanation and a new question to test that diagnosis.

Evaluation: how to know the models work

- Individual model: per-class precision, recall and F1; confusion matrix; performance on unseen question forms.
- Sequence model: per-pattern precision/recall/F1; test on unseen sequences and report how often it abstains when evidence is weak.
- Intervention: teacher review for correctness and relevance; check whether students improve on new, related questions.
- End-to-end: measure diagnosis accuracy and the proportion of students who demonstrate improvement after intervention. A correct explanation alone is not proof that learning improved.

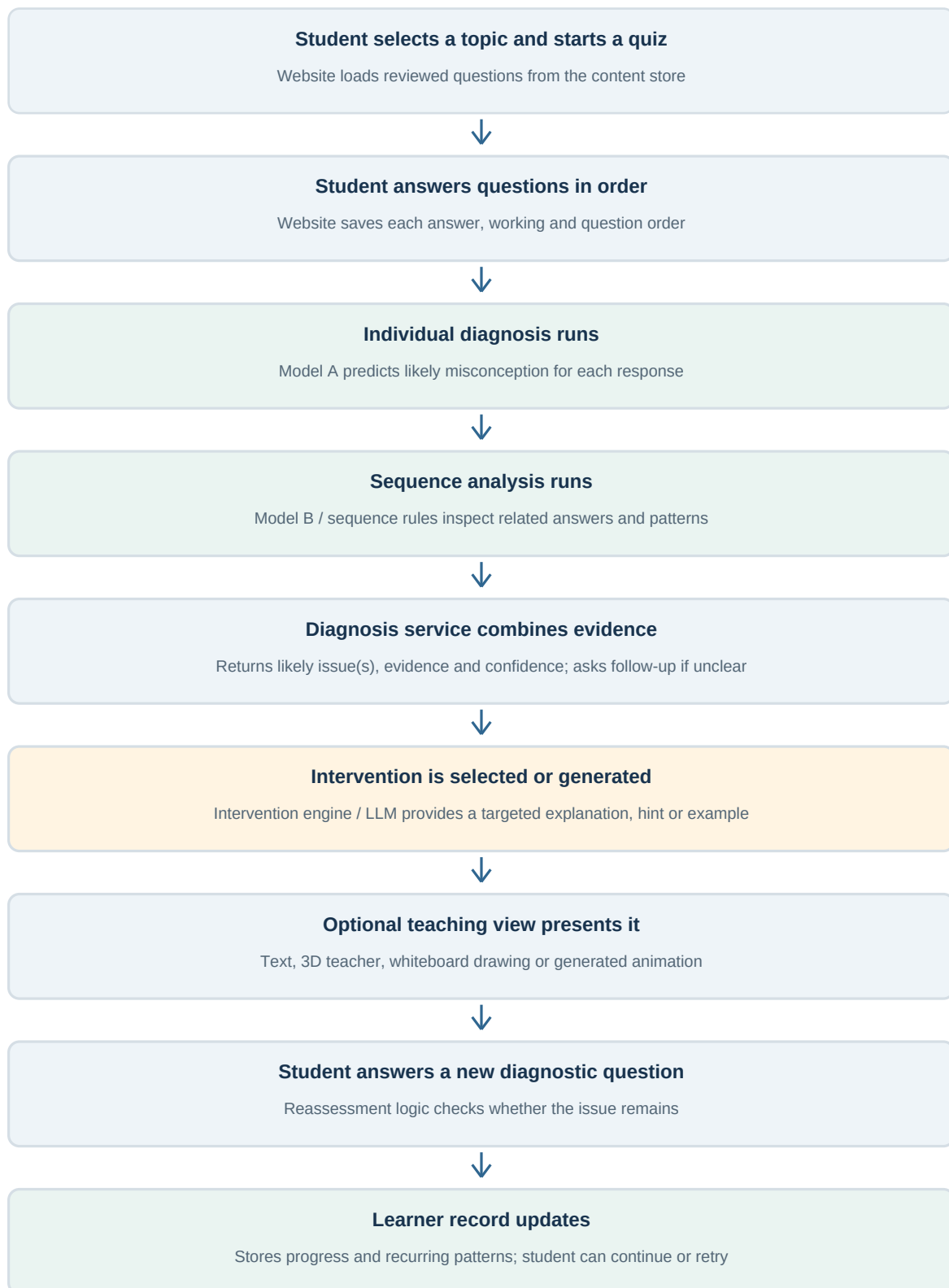
Part 3 — Website and integration architecture

The website is a demonstration interface, not the main research contribution. Keep it small: one subject, a few topics, a quiz, a diagnosis result, an intervention and reassessment.

Layer architecture

Layer	What it does	Main features
1. Student UI	Displays the experience and collects input.	Choose topic, take quiz, enter answer/work, view explanation, answer follow-up and see progress.
2. API / session service	Connects the UI to backend services and stores attempts.	Create quiz session, save ordered responses, fetch diagnosis and intervention.
3. Content and question store	Holds reviewed educational content and labels.	Topics, questions, correct solutions, misconception taxonomy, follow-up questions and intervention resources.
4. Diagnosis service	Runs the trained components.	Individual classifier + sequence model/rules; combines evidence and confidence.
5. Intervention service	Chooses the next teaching action.	Selects a curated resource or calls an LLM to produce a targeted explanation, hint or question.
6. Reassessment and learner record	Checks progress and keeps useful history.	New question, new response analysis, unresolved/resolved/uncertain status and misconception history.
7. Optional presentation tools	Makes the intervention visual or spoken.	3D avatar, text-to-speech, whiteboard renderer and animation tool.

Clean user flow — with models shown where they run



Where the teaching part fits

A short lesson before the quiz may be included to make the demo feel like an LMS, but it is optional. The required teaching action is the **targeted intervention after diagnosis**. The 3D character and whiteboard are presentation options for that intervention; they do not replace the diagnosis or reassessment.

Part 4 — Optional features and how to integrate them

These features can make the demo easier to understand and more engaging. Add them after the core diagnosis flow works.

Optional feature	How it works	Where it connects	Practical first version
3D AI teacher	Turns the intervention text into speech; an avatar plays audio with lip-sync and simple gestures.	After the intervention engine returns an explanation.	Use one ready-made avatar, text-to-speech and basic lip-sync. Avoid building a custom 3D model pipeline first.
AI whiteboard	Converts an explanation into ordered drawing commands: write text, draw a line, plot a graph, show a formula or diagram.	Intervention output includes a structured visual plan; frontend whiteboard renders it.	Support a small set of Physics primitives: axes, arrows, labels, equations and simple diagrams. Animate strokes in sequence.
Concept animation	Shows a motion or process over time, such as a moving object or changing force diagram.	Intervention engine requests an animation template based on the misconception.	Use parameterised, prebuilt animations rather than asking a generative model to create arbitrary video live.
Interactive simulation	Lets a student change a value and observe the result.	Can be offered as an intervention activity or a follow-up exploration.	Build one focused simulation, such as changing distance/time or adjusting an inclined-plane angle.
Pre-quiz chapter lesson	Shows a brief explanation before the diagnostic quiz.	At the beginning of the website flow; separate from misconception diagnosis.	Use a small amount of reviewed content for the selected topic.

Whiteboard output should be structured

Do not rely on a language model to directly draw pixels. Ask it for a limited set of drawing instructions, validate them, and let the website render them. For example:

Instruction	Rendered result
<code>draw_axes(x_label='time', y_label='distance')</code>	Coordinate axes with labels
<code>plot_line(points=[[0,0),(5,100)])</code>	A distance-time line graph
<code>write_equation('speed = distance / time')</code>	Formula appears on the board
<code>highlight('divide distance by time')</code>	Key step is highlighted

Safety and quality checks for optional generation

- Use reviewed Physics facts and formulas; validate generated equations and diagrams.
- Keep generated drawing commands within an allowed format. Do not execute arbitrary code returned by a model.
- Keep the visual explanation aligned with the diagnosed misconception. A flashy animation that does not address the error adds little value.

Part 5 — Team split and build order

Suggested split for four teammates

Owner	Main responsibility	Deliverable
Teammate 1 — Dataset	Choose a small Physics scope, write questions, define misconception labels, create individual and sequence examples, arrange expert review.	Versioned, labelled dataset + taxonomy + data guide.
Teammate 2 — Individual model	Build baseline classifier, evaluate per misconception, add confidence/unknown handling.	Runnable individual diagnosis API or module + evaluation results.
Teammate 3 — Sequence model	Represent ordered quiz attempts, build pattern rules/baseline sequence classifier, evaluate sequence-level predictions.	Sequence analysis module + test results.
Teammate 4 — Website/integration	Build quiz flow, backend endpoints, combine model outputs, display intervention and reassessment. Coordinate with the team for model interfaces.	Working end-to-end demo and integration contract.

Shared responsibility: intervention and content

All teammates should agree on the misconception labels and output format. One person can connect an existing LLM or a curated intervention bank after the diagnosis modules are available. Optional avatar/whiteboard work should be a separate stretch task, not a blocker for the core demo.

Recommended build order

- **Step 1:** Select one Physics chapter and a manageable set of misconceptions.
- **Step 2:** Create and review the individual dataset and a smaller sequence dataset.
- **Step 3:** Build a rule-based baseline, then train and evaluate the individual classifier.
- **Step 4:** Build sequence analysis and compare it against individual-only diagnosis.
- **Step 5:** Agree on a shared prediction format, for example: misconception ID, confidence, evidence and whether more information is needed.
- **Step 6:** Connect the intervention engine and a new question for reassessment.
- **Step 7:** Build the website flow and demonstrate the complete cycle.
- **Step 8:** Add one optional visual feature only if the core system is stable.

Suggested shared output format

The individual and sequence modules should return consistent fields. Example (illustrative only):

```
student_attempt_id: "A-014"  
predicted_misconception: "speed_time_operation"  
confidence: 0.78  
evidence: ["Q2: multiplied distance and time", "Q5: repeated same operation"]  
status: "likely"  
next_action: "targeted_explanation_and_check"
```

Part 6 — Scope, risks and references

What is essential versus optional?

Core to the problem	Optional demo enhancements
• Labelled misconception dataset • Individual diagnosis • Analysis of answer patterns across related questions • Personalised intervention • Reassessment and progress tracking • Evaluation on unseen examples	• Full chapter-by-chapter LMS • 3D avatar and lip-sync • Live voice conversation • Animated whiteboard • AI-generated animations/video • Interactive Physics simulations • Large content library

Common mistakes to avoid

- Do not label every wrong answer as a misconception. Separate conceptual errors from arithmetic slips, unit errors, guessing and missing evidence.
- Do not claim a broad misconception from one response unless the evidence supports it. Use a follow-up question when needed.
- Do not train and test on near-identical copies of the same question. That can make accuracy look higher than real-world performance.
- Do not make the LLM the only evaluator. Track measurable model performance and learning outcomes.
- Do not spend most of the project time building a general LMS or avatar before the diagnosis and reassessment loop works.

Useful references and starting points

These are general technical resources for the methods proposed here; they are not claims about extra requirements in the problem statement.

- [Scikit-learn: supervised learning](#) — baseline classifiers and evaluation workflows.
- [Hugging Face Transformers documentation](#) — sequence and language-model tooling if a neural approach is justified.
- [TensorFlow](#) — model training and deployment options.
- [Prompt engineering guide](#) — structuring LLM outputs for controlled explanations and interventions.
- [MDN Canvas API](#) — browser drawing surface for a custom whiteboard.

Final summary: Build the dataset first. Train and evaluate an individual misconception model and a sequence/pattern component. Combine their evidence, generate a targeted intervention, and reassess with a new question. Use a small LMS-style website to demonstrate this cycle. Treat the 3D teacher, whiteboard and animations as optional ways to present the intervention—not as the core AI solution.

Requirement coverage check

This appendix maps the original Re:Learn problem statement to the project documentation. The items below are the requirements your final prototype and evaluation should demonstrate—not merely mention.

Problem-statement requirement	Where it is covered / what to demonstrate	Status
Choose one domain: programming, algebra or physics	Documentation selects Physics. Keep the prototype within one chosen domain and state the exact chapter/topic scope.	Covered; confirm scope
Analyse learner answers, working or code	Dataset and answer-analysis sections include responses, context and working. For Physics, support numerical steps or written reasoning where available.	Covered
Identify the underlying misconception, not just wrong/correct	Individual misconception classifier and labelled dataset are specified. Output should name a likely misconception or return uncertain.	Covered
Distinguish different misconceptions that produce similar wrong answers	Dataset design, individual model, diagnostic follow-up and evaluation address this. Include deliberately similar-error examples in testing.	Covered; test explicitly
Use diagnosis to generate or select a targeted intervention	Intervention engine uses the diagnosis to select a resource or generate a targeted explanation, hint or exercise.	Covered
Reassess after intervention	Reassessment asks a new, related question and analyses the response after the intervention.	Covered
Determine whether the misconception is actually resolved; do not assume one correct follow-up proves learning	Make the resolution rule explicit: use new evidence across suitable questions/attempts; allow unresolved or uncertain; avoid declaring mastery from one answer alone.	Clarification added
Track recurring misconceptions and demonstrated understanding across attempts	Learner model stores misconception history, attempts and evidence of understanding. Track both remaining errors and successful transfer to new questions.	Covered; implement both
Evaluate diagnosis accuracy and performance on unseen responses or misconceptions	Use held-out tests. Report standard classification metrics and a separate generalisation test on response forms and, where feasible, misconception categories excluded from training.	Clarification added
Build a misconception dataset of correct/incorrect responses and underlying misconceptions	Dataset section defines question, correct solution, student response, misconception label, evidence and sequence fields.	Covered

Conclusion: The original documentation contains a plan for every core requirement. The most important distinction is that sequence/pattern analysis is an additional design choice from your team discussion; the problem statement does not explicitly require a separate sequence model. It can strengthen the learner model and diagnosis, but it should not delay the required individual diagnosis, intervention and resolution assessment.

Team to-do page

Use this as the project checklist. Assign an owner and mark each item only when its output has been reviewed or tested.

A. Scope and misconception map

- Confirm Physics as the selected domain and choose a manageable chapter/topic scope.
- List the concepts and define a clear misconception taxonomy for each concept.
- Include common conceptual errors plus arithmetic slips, unit errors, guesses, unknown and insufficient-evidence cases.
- Identify pairs of misconceptions that can produce similar answers; design questions that distinguish them.

B. Dataset creation and review

- Use a permitted textbook/source for topic alignment; write or adapt questions with usage rights in mind.
- Create individual-response records: question, context, correct answer/steps, wrong response/working and reviewed misconception label.
- Create ordered quiz sequences with sequence IDs, question order, responses and reviewed pattern-level interpretations.
- Generate realistic wrong-answer examples from the reasoning behind each misconception—not just random incorrect values.
- Have a subject reviewer check questions, correct solutions, labels and intervention content.
- Version the dataset and document label definitions, uncertain cases and data splits.

C. Individual diagnosis model

- Build a simple baseline classifier or transparent rule-based baseline.
- Train the individual misconception model using the reviewed individual-response dataset.
- Return misconception label(s), confidence and an unknown/needs-more-evidence option.
- Evaluate per-class precision, recall, F1 and confusion matrix.
- Test on held-out response forms/question templates, including similar-error cases.

D. Sequence/pattern analysis (team extension)

- Define what counts as a related question sequence and which patterns the system should detect.
- Build a sequence baseline using rules/features before trying a larger neural model.
- Train/evaluate a sequence model only if the labelled sequence data is sufficient.
- Compare sequence-level results with individual-only diagnosis; record when sequence evidence changes the diagnosis.
- Do not treat answer patterns as proof; return uncertainty or ask a diagnostic follow-up when needed.

E. Intervention and resolution assessment

- Define an intervention format: diagnosis, explanation/hint, worked example and targeted follow-up question.
- Connect a curated intervention bank or controlled LLM generation to the diagnosis output.
- Review generated explanations and verify Physics correctness.
- Prepare new, related reassessment questions that test the same concept in a different way.
- Define explicit outcomes: likely resolved, still present, or uncertain; do not equate one correct answer with resolution.
- Store recurring misconceptions and evidence of demonstrated understanding across attempts.

F. Website and integration

- Build the minimum demo flow: select topic → quiz → diagnosis → intervention → reassessment → progress.
- Create a shared interface between the website, individual model, sequence component and intervention service.
- Save answer order, working where available, model outputs, confidence and reassessment results.
- Show the diagnosis and a simple reason/evidence summary without overstating certainty.
- Test the complete flow with several prepared student-answer scenarios.

G. Evaluation and final demo

- Evaluate diagnosis accuracy on unseen responses and clearly state which data was held out.
- Where feasible, hold out one or more misconception types to test generalisation to unseen misconceptions; report limitations if data is too small.
- Measure intervention quality and whether learners improve on new related questions.
- Demonstrate at least one case where two similar wrong answers have different causes.
- Demonstrate a sequence where multiple answers reveal a recurring or connected misconception.
- Document failures, uncertain cases, metrics and what the prototype does not yet support.

H. Optional presentation features — only after core flow

- Choose at most one stretch feature first: 3D teacher, animated whiteboard, concept animation or simulation.
- If using a 3D teacher, connect intervention text to speech and basic avatar lip-sync.
- If using a whiteboard, define safe structured drawing commands and render them in the frontend.
- Validate formulas, diagrams and generated visuals; keep the visual intervention tied to the diagnosis.

Definition of done for the core prototype: A student can submit answers; the system can identify a likely misconception with evidence or express uncertainty; a targeted intervention is delivered; a new question is used to reassess; and the system records whether the misconception appears resolved, remains present or is uncertain. The team can show evaluation results, including a test on responses not used during training.

Re:Learn — Dataset Update

Supplement to the technical project documentation | Question formats, dataset evidence, and collection workflow

1. Expanded question formats for Class 10 Physics

The project is not limited to multiple-choice questions. Use different formats when they help reveal the reasoning behind an answer. A response may include a selected option, a numerical result, written explanation, calculation steps, a graph interpretation, or a description of a diagram.

Format	What it can test	Useful evidence to retain
MCQ / select an option	Common conceptual distinctions and distractor choices	Chosen option and, where possible, a brief reason
Short conceptual / theory answer	Definitions, causal explanations, and conceptual models	Exact response text and key reasoning phrases
Numerical problem	Formula choice, substitution, arithmetic, and units	Intermediate steps, formula used, final value, units
Assertion–reason	Whether a learner understands a claim and its explanation	Both selections and the stated relationship between them
Case / application-based	Transfer of a concept to a real-world or unfamiliar context	Answer plus the concept applied to the scenario
Diagram / circuit / ray-diagram interpretation	Spatial or visual understanding of a physical setup	Selected labels, drawn changes, or a description of the interpretation
Graph / table interpretation	Relationships, trends, slope, axes, and units	Read values, explanation, graph markings, and units
Error analysis / explain a worked solution	Ability to locate and explain an incorrect step	Identified step and the learner's explanation of the correction

For an initial prototype, prioritize MCQs, short conceptual answers, and numerical questions with working. Add diagram, graph, assertion–reason, or case-based formats only when the team can capture and evaluate the response reliably. Do not claim image understanding if the system only receives text.

2. Dataset features and evidence

The dataset should help the system infer why an answer was given, not merely map a question to its official answer. Preserve enough context to distinguish misconceptions that produce similar outcomes.

Feature	Why it matters
Question wording, format, topic, and concept	Provides the context in which the response was produced
Correct answer and accepted solution paths	Allows valid alternative methods to be recognized
Student response and intermediate work	Shows the evidence behind a possible diagnosis
Misconception hypothesis and supporting evidence	Separates an observed error from an inferred underlying cause
Alternative cause, uncertainty, or insufficient evidence	Prevents forced or overconfident labeling when evidence is ambiguous
Response origin: real, sourced, authored, or synthetic	Makes the provenance and limitations of examples explicit
Source, chapter/page or paper identifier, and usage notes	Supports traceability, review, and rights checks
Question/response family and duplicate relationship	Helps avoid near-duplicate leakage across evaluation splits
Session/attempt order, when available	Supports optional tracking of recurring patterns over time

Feature	Why it matters
Intervention and reassessment outcome, when available	Connects diagnosis to whether understanding changes after support

Keep sequence-level analysis as a related extension. The core requirement can be demonstrated with individual-response diagnosis, targeted intervention, and reassessment; longitudinal patterns add value when reliable ordered responses are available.

3. Resource collection and quality controls

- Choose one primary Class 10 Physics textbook as the authoritative syllabus reference. Use other resources to supplement questions or investigate misconceptions, not to silently mix competing syllabi.
- Search existing datasets, textbook materials, examination papers, sample papers, and answer keys before writing large numbers of examples.
- Prefer resources with worked solutions or student reasoning. An answer key alone usually cannot establish why a student made an error.
- Record where material came from and check its access and reuse terms. Do not bypass access restrictions or treat a publicly viewable resource as automatically reusable.
- Keep extracted source material, team-authored examples, and AI-generated synthetic responses distinguishable.
- Review question validity, answer correctness, misconception interpretation, and intervention relevance with a subject-aware reviewer.
- Include correct responses, plausible incorrect responses, calculation slips, unit errors, unclear responses, and unknown/insufficient-evidence cases.
- Test generalization using held-out question templates or response families. Avoid splitting near-identical variants between training and test data.

Important limitation: an incorrect final answer is not sufficient proof of a particular misconception. When working or explanation is absent, return uncertainty or ask a diagnostic follow-up question.

4. Updated practical workflow

1. Select the primary textbook and define a manageable set of Class 10 Physics concepts.
2. Find relevant existing datasets and source materials; document what each resource actually contains.
3. Collect questions and verified answers, then create or collect varied response examples with clear provenance.
4. Have likely misconception labels and ambiguous cases reviewed rather than inferred from answer keys alone.
5. Build the individual-response baseline first. Add session-level patterns only when ordered, trustworthy data exists.
6. Link each supported misconception to a reviewed intervention and a fresh reassessment question.
7. Evaluate diagnosis quality, uncertainty handling, performance on unseen question forms, and whether reassessment provides evidence of improvement.